Project Title:

Invoice Management System

Who is the business/client, and what is the one goal?

Target users:

Freelance business owners, initially focusing on freelance software developers who deliver complete technology projects such as websites, applications, information systems, AI models, and other technology solutions. Their clients may be individuals, professional groups, or companies/corporates.

Current problem:

Invoice and contract information are managed manually, particularly through Excel/spreadsheets, while payment reminders are also handled manually. This makes it harder to keep financial information centralized, remember payment follow-ups, and obtain organized financial information when needed.

System Goal:

Provide a centralized system for managing customers, contracts, invoices, payments, and reminders, while automating payment reminders to both the freelancer and the customer and providing analytical reports for a selected period, such as monthly or yearly reporting.

1. Business and primary goal:

- **Target business:** Freelance businesses, initially focused on freelance software developers.
- **Services:** Complete technology projects such as websites, applications, systems, AI models, and other technology solutions.
- **Clients:** Individuals, professional groups, and corporations.
- **Current approach:** Invoice and contract information are managed manually, particularly through Excel/spreadsheets, and payment reminders are handled manually.

- **Problem:** Information is not centralized; follow-ups can be forgotten, reminders require manual work, and obtaining organized financial/analytical information for a specific period is inconvenient.
- **System purpose:** Centralize customer, contract, invoice, payment, and reminder management; automate invoice/payment reminders; and provide analytical reports for selected periods.

Functional Requirements:

Module 1 — Authentication:

FR-001
The system shall allow the administrator to log in using a registered email address and password.

FR-002
The system shall reject the login attempts and display a single “invalid email or password” message, without indicating which one is incorrect.

FR-003
The system shall temporarily lock the administrator account after five consecutive failed login attempts for fifteen minutes.

FR-004 ⚠️
The system must invalidate the current session and redirect to the login page when the administrator logs out, so that the previous session token no longer grants access.

FR-005 ⚠️
The system must redirect any unauthenticated request for a page other than the login page to the login page.

Module 2 — Dashboard:

FR-006

The system must display the number of customers that are not archived on the dashboard.

FR-007

The system must display the sum of remaining balances across all issued invoices that are not fully paid and not cancelled.

FR-008

The system shall display the total amount collected during the current month (30 days).

FR-009 ⚠️

The system must display all invoices whose due date has passed and whose remaining balance is greater than zero, sorted by days overdue, on the dashboard

FR-010

The system shall display payments (invoices) due within the next seven days.

FR-011 ⚠️

The system must display the 20 most recent events (invoice issued, payment recorded, reminder sent, customer created) with a timestamp and a link to the related record, ordered newest first.

Module 3 — Customer Management:

FR-012 ⚠️

The system must create a customer record with name, email, phone, and billing address when the administrator submits the customer form, rejecting the

submission with a field-level message when name or email is missing, or the email is already registered.

FR-013 ⚠️

The system must allow the administrator to update a customer's name, email, phone, and billing address, applying the same validation rules as creation.

FR-014 ⚠️

The system must exclude an archived customer from customer selection lists and from the active customer count, while keeping their existing invoices and payments intact and viewable.

FR-015

The system shall prevent archiving a customer who has active unpaid invoices.

FR-016

The system must display all customers whose name contains the entered text, matched case-insensitively, when the administrator searches by name.

FR-017

the system must allow filtering customers by status, where status is one of active or archived

FR-018

The system shall display all invoices associated with a customer.

Module 4 — Contract Management:

FR-019 ⚠️

The system must create a contract for a selected customer with a title, start date, expiration date, contract value, and billing frequency when the administrator submits the contract form.

FR-020

The system must allow the administrator to edit a contract title, expiration date, and value while its expiration date is in the future and must reject edits to an expired contract.

FR-021

The system must exclude an archived contract from active contract lists and from expiry notifications, while keeping it viewable on the customer's record.

FR-022
The system shall associate each contract with exactly one customer.

FR-025

The system must send the administrator an email notification 30 days and 7 days before a contract expiration date and must record that the notification was sent.

Module 5 — Invoice Management:

FR-026 ⚠️

The system must create a draft invoice for a selected customer with a due date and one or more-line items (description, quantity, and unit price) when the administrator submits the invoice form.

FR-027 ⚠️
The system must issue a draft invoice when the administrator confirms it, assigning the next number from the configured invoice sequence, setting the issue date, and locking the invoice number and line items.

FR-028 ⚠️
The system must allow the administrator to edit a draft invoice, including its due date, line items, discount, and applicable tax settings.

FR-029

The system shall prevent modification of fully paid invoices.

FR-030

The system shall allow the administrator to cancel an issued invoice that has not been fully paid. When an invoice is cancelled, its existing payment history must be preserved, and the invoice must no longer be included in outstanding balances or reminder processing.

FR-031 ⚠️

The system must calculate the invoice total as the sum of its line items (quantity x unit price), minus any discount, plus applicable tax, and must recalculate whenever a line item, discount, or tax rate changes.

FR-032

The system must apply the configured tax rate to the invoice subtotal after any discount and must store the tax amount and the rate used on the invoice at the time it is issued.

FR-033

The system must allow a discount to be applied to an invoice as either a percentage of the subtotal or a fixed amount and must apply it to the subtotal before tax is calculated.

FR-034

The system shall associate each invoice with one customer.

FR-035

The system must generate a PDF of an issued invoice containing the business details, customer details, invoice number, issue and due dates, line items, subtotal, discount, tax, total, and remaining balance

FR-036: The system must allow the administrator to send an issued invoice to the

associated customer by email, including the invoice PDF as an attachment, using the customer's registered email address.

FR-037: The system must record the date, time, and delivery status of each invoice sent to the customer.

Module 6 — Installment Management:

FR-040 ⚠️

The system must allow the administrator to divide an issued invoice into two or more installments, each with an amount and a due date, and must reject the schedule when the installment amounts do not sum exactly to the invoice total.

FR-043 ⚠️

The system must display each installment with its amount, due date, amount paid, and status (unpaid, partially paid, paid, or overdue).

FR-042 ⚠️

The system must allocate a recorded payment to the oldest unpaid installment first, carrying any remainder to the next installment.

Module 7 — Payment Management:

FR-046 ⚠️

The system must record a payment against a specified invoice with an amount, a payment date, and a payment method (cash, bank transfer, cheque, or card) when the administrator submits the payment form.

FR-047

The system shall allow partial payments.

FR-048 ⚠️

The system must recalculate an invoice's remaining balance as the total minus the sum of payments recorded against it and must reject a payment whose amount exceeds the current remaining balance.

FR-049

The system shall mark an invoice as fully paid once its remaining balance reaches zero.

FR-050

The system shall maintain the complete payment history for each invoice.

FR-051

The system shall calculate and display an invoice's remaining balance as the invoice total minus the sum of all recorded payments.

Module 8 — Reminder System:

FR-051

The system must allow the administrator to configure the reminder's schedule as a set of offsets relative to the due date, defaulting 7 days before, on the due date, and 3, 14, and 30 days after.

FR-052 ⚠️

The system must send the customer an email reminder with the invoice PDF and the amount due 7 days before the invoice due date, when the invoice is not fully paid.

FR-053 ⚠️

The system must send the customer an email reminder on each configured overdue offset while the invoice remains unpaid and must show the invoice in the administrator's overdue list.

FR-054

The system shall record the reminder's history for each invoice.

FR-055

The system shall prevent duplicate reminders from being generated on the same scheduled date**.**

Module 9 — Reports:

FR-056

The system must display, for a selected month, the total amount invoiced and the total amount collected, as two separate figures.

FR-057

The system must display each customer's total outstanding balance, sorted highest first.

FR-058

The system shall provide analytical reports for a user-selected period, including total invoiced amount, total payments received, total outstanding amount, total overdue amount, and invoice status summaries.

FR-059

The system shall display payments received during a selected period.

FR-061 ⚠️

The system must export the selected report as a CSV file containing the same rows and columns shown on screen.

Module 10 — Search:

FR-062

The system shall allow searching invoices by invoice number.

FR-063

The system shall allow searching invoices by customer.

FR-064 ⚠️

The system must allow filtering invoices by status, where status is one of draft, issued, partially paid, paid, overdue, or cancelled.

FR-065 ⚠️

The system must allow filtering invoices by an issue-date range with a start and end date.

Module 11 — Settings:

FR-066

The system must allow the administrator to update the business name, address, phone, email, tax registration number, and logo, and must use these values on generated invoice PDFs.

FR-067

The system must allow the administrator to configure an invoice number to prefix and starting sequence and must assign numbers sequentially without reusing a number from a cancelled invoice.

FR-068

The system must allow the administrator to configure the default tax rate, applying it only to invoices issued after the change.

FR-069: The system shall allow the administrator to configure which critical events trigger email notifications, including overdue invoices, significantly overdue invoices, and received payments.

FR-070: The system shall allow the administrator to customize the subject and body of administrator notification email templates using the available invoice and customer information.

Module 12 — CSV: (This module is in version 2 – don't consider it for now)

Invoice CSV

FR-075
The system shall allow the administrator to import invoice records from a CSV file.

FR-076
The system shall validate invoice fields, customer references, dates, amounts, and payment status when importing a CSV file.

FR-077
The system shall allow the administrator to export invoice records as a CSV file.

Payment CSV

FR-079
The system shall allow the administrator to import payment records from a CSV file.

FR-080
The system shall validate payment amounts, payment dates, payment methods, and associated invoice references when importing a CSV file.

FR-081
The system shall allow the administrator to export payment records as a CSV file.

Customer CSV Handling

FR-083
The system shall allow the administrator to import customer records from a CSV file.

FR-084
The system shall validate required customer fields when importing a CSV file.

FR-085
The system shall allow the administrator to export customer records as a CSV file.

Non-Functional Requirements:

Performance:

NFR-PERF-001

The system shall return responses for standard API requests within 500 milliseconds **for** at least 95% of requests under normal operating conditions.

NFR-PERF-002

The system shall load the main dashboard within 2 seconds for at least 95% of page loads under normal operating conditions.

Reliability and Availability:

NFR-REL-001

The system must maintain at least 99% availability measured monthly, excluding scheduled maintenance announced 24 hours in advance.

NFR-REL-002

The system must retain every committed invoice, payment, customer, and contract record after an unexpected process termination, verified by killing the application mid-transaction, restarting, and confirming no committed record is lost or duplicated.

NFR-REL-003

The system must return an error page with a plain-language message and a reference code, and must not expose stack traces, SQL, or connection strings, when the database or an external service is unavailable.

Security:

NFR-SEC-001

The system must store administrator passwords hashed with bcrypt at a cost factor of at least 12 and must never store or log a password in plaintext.

NFR-SEC-002

The system shall terminate an authenticated session after 30 minutes of inactivity.

Scalability:

NFR-SCAL-001

The system must meet the NFR-PERF targets with 500 customers, 10,000 invoices, and 25,000-line items stored.

NFR-SCAL-002

The system must meet the NFR-PERF targets with 50,000 payment records stored.

Usability:

NFR-USA-001

The system must allow an administrator to create a customer and issue an invoice to them in under 3 minutes and no more than 12 clicks, verified with two people who have not used the system before.

NFR-USA-002

The system shall display validation errors next to the relevant input fields and provide a clear description of the required correction.

Maintainability:

NFR-MNT-001

The system shall maintain automated tests covering at least 70% of the backend business-logic code.

Compatibility:

NFR-COMP-001

The system must function correctly on the latest two major versions of Chrome, Edge, Firefox, and Safari.

NFR-COMP-002

The system must render all pages without horizontal scrolling and with all controls reachable at viewport widths of 375px, 768px, and 1440px.

Accessibility:

NFR-ACC-001

The system must meet WCAG 2.1 Level AA for name, role, and value on all interactive controls, verified with an automated accessibility scan reporting zero violations of that rule.

NFR-ACC-002

The system must associate each validation message with its input programmatically, so that a screen reader announces the error when focus moves to the field.

Pending Issues Resolution in V1:

Invoice:

1. decide how the system treats a draft invoice.

   A draft invoice should be able to exist **without consuming a final invoice number**.
   Draft creation
   → customer + dates + line items
   → status = `draft`
   → no final invoice number yet

   Issue
   → assign next invoice number
   → lock the number
   → lock the line items
   → set/finalize issue date
   → status = `issued`

2. issue date:
   We need to distinguish:
   Draft creation date — when the draft was created.

Issue date — when the invoice becomes an official issued invoice.

For the invoice itself, I think the **issue date belongs to the issue of action**, not the initial draft creation.

Stage	Number	Line items	Issue date	Status
Draft	Not final/assigned	Editable	Not finalized	Draft
Issued	Assigned	Locked	Set	Issued
Partially paid	Assigned	Locked	Set	Partially paid
Paid	Assigned	Locked	Set	Paid
Overdue	Assigned	Locked	Set	Overdue
Cancelled	Assigned if issued	Locked	Set	Cancelled

3. Modification of a fully paid invoice:

we keep FR-029, but the overall rules should make it clear that issuing already locks the financial content.

4. Cancellation:

keeping the payment recorded and not implementing refunds in v1, unless the freelancer specifically needs refund management.

If an issued invoice has been partially paid and is then cancelled, the recorded payment remains associated with the cancelled invoice. The payment history is preserved, and refund management is outside v1.

5. Discount:
rejects a discount greater than the subtotal, otherwise the calculation could produce a negative taxable amount.

How does the invoice reach the customer?

Email: v1

The system can potentially:

- generate the PDF
- attach it
- send it to the customer's email
- record that it was sent

WhatsApp — v2

Reminders:

Invoice issued

↓

PDF generated

↓

Invoice emailed to customer

↓

7 days before due date → reminder email

↓

Due date → reminder email

↓

3 days overdue → reminder email

↓

14 days overdue → reminder email

↓

30 days overdue → reminder email

↓

Payment received at any point → stop future reminders

What about reminders to the freelancer?

Customer: automatic email reminders.

Administrator: dashboard/overdue list + possibly email notifications.

Critical Notification for admin:

Invoice reaches 30 days overdue

Payment is received

Payment fails

Administrator Notification Settings:

Notification	Email?
Invoice overdue	ON
Invoice 30+ days overdue	ON
Payment received	ON/OFF
Contract expiring	ON
Other routine events	OFF

Invoice status:

Draft

Invoice has been created but not issued.

Issued

Invoice has been issued, has no payment recorded, and is not overdue.

Partially Paid

At least one payment has been recorded, but the remaining balance is greater than zero, and the invoice is not overdue.

Overdue

The remaining balance is greater than zero and the due date has passed.

Paid

Remain balance = zero.

Cancelled

The administrator has cancelled the invoice.

Why does the system have contracts, and what role do contracts play in the invoice/payment workflow?

The contract represents the agreement between the freelancer and the customer, while invoices represent the actual amounts billed under that agreement.

Contracts allow the administrator to record and manage agreements with customers and associate related invoices with the appropriate contract for reference and organization.

Contract as a record/document

The contract is basically a place to store:

- contract number

- client
- project/service description
- start/end dates
- status
- agreement information

Then invoices can be linked to it simply for organization and reference.

Flow:

Customer → Contract → Invoice → Payment

Reports

Provides **historical and analytical information for a selected period**.

For example:

- total invoiced amount
- total payments received
- total outstanding amount
- total overdue amount
- number of invoices issued
- number of paid/unpaid/overdue invoices
- customer-based financial information
- monthly/yearly/custom-period analysis

Reports provide historical and analytical financial information for a user-selected period, while the Dashboard provides a current operational overview of the system.